

Ex. No. : 06

Date :

Register No. : 221701701

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SD Card

Aim

Implement an application to write the Register Number, Name and CGPA to the SD card in text file format.

Procedure:

Step 1 : File → New Project

Provide the application name (e.g., "SD Card") and click "Next".

Step 2 : Select the target Android devices

Select the minimum SDK to run the application. Click "Next".

Step 3 : Choose the activity for the application

By default, choose "Blank Activity". Click "Next".

Step 4 : Enter activity name and click "Finish".

Step 5 : Edit the program

Request storage permissions in the manifest file.

Use file input/output streams in MainActivity.kt to read from and write to SD card or internal storage.

Step 6 : Run the application

Two ways to run the application:

1. Running through emulator (with SD card configured)
2. Running through mobile device

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.SdCard"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="SD Card"
        android:textSize="24sp"
        android:textStyle="bold"
        android:textColor="@android:color/white"
        android:background="#6200EE"
        android:padding="12dp"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:layout_marginTop="16dp">

        <EditText
            android:id="@+id/editTextRegisterNumber"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter the register number..."
            android:inputType="text"
            android:padding="12dp"
            android:background="@drawable/edit_text_background"
```

```
android:layout_marginBottom="8dp"/>
```

```
<EditText
```

```
    android:id="@+id/editTextName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter the name..."
    android:inputType="textPersonName"
    android:padding="12dp"
    android:background="@drawable/edit_text_background"
    android:layout_marginBottom="8dp"/>
```

```
<EditText
```

```
    android:id="@+id/editTextCGPA"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter the CGPA..."
    android:inputType="numberDecimal"
    android:padding="12dp"
    android:background="@drawable/edit_text_background"
    android:layout_marginBottom="16dp"/>
```

```
<Button
```

```
    android:id="@+id/buttonSave"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Save"
    android:textColor="@android:color/white"
    android:backgroundTint="#6200EE"
    android:padding="12dp"
    android:layout_marginBottom="8dp"/>
```

```
<Button
```

```
    android:id="@+id/buttonLoad"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Load"
```

```
        android:textColor="@android:color/white"
        android:backgroundTint="#6200EE"
        android:padding="12dp"/>
    </LinearLayout>

</LinearLayout>
```

MainActivity.kt

```
package com.example.sdcard

import android.Manifest
import android.content.pm.PackageManager
import android.os.Bundle
import android.os.Environment
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import androidx.core.app.ActivityCompat
import androidx.core.content.ContextCompat
import java.io.File
import java.io.FileOutputStream
import java.io.IOException

class MainActivity : AppCompatActivity() {
    private lateinit var registerNumberEditText: EditText
    private lateinit var nameEditText: EditText
    private lateinit var cgpaEditText: EditText
    private lateinit var saveButton: Button
    private lateinit var loadButton: Button
```

```

private val STORAGE_PERMISSION_CODE = 100

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)

    // Initialize UI components
    registerNumberEditText = findViewById(R.id.editTextRegisterNumber)
    nameEditText = findViewById(R.id.editTextName)
    cgpaEditText = findViewById(R.id.editTextCGPA)
    saveButton = findViewById(R.id.buttonSave)
    loadButton = findViewById(R.id.buttonLoad)

    // Request storage permissions
    requestStoragePermission()

    // Set click listeners
    saveButton.setOnClickListener {
        saveDataToSD()
    }

    loadButton.setOnClickListener {
        loadDataFromSD()
    }
}

private fun requestStoragePermission() {
    if (ContextCompat.checkSelfPermission(
        this,
        Manifest.permission.WRITE_EXTERNAL_STORAGE
    )

```

```

        ) != PackageManager.PERMISSION_GRANTED
    ) {
        ActivityCompat.requestPermissions(
            this,
            arrayOf(
                Manifest.permission.WRITE_EXTERNAL_STORAGE,
                Manifest.permission.READ_EXTERNAL_STORAGE
            ),
            STORAGE_PERMISSION_CODE
        )
    }
}

private fun saveDataToSD() {
    // Validate inputs

    val registerNumber = registerNumberEditText.text.toString().trim()
    val name = nameEditText.text.toString().trim()
    val cgpa = cgpaEditText.text.toString().trim()

    if (registerNumber.isEmpty() || name.isEmpty() || cgpa.isEmpty()) {
        Toast.makeText(this, "Please fill all fields",
            Toast.LENGTH_SHORT).show()
        return
    }

    try {
        // Try multiple storage options for emulator compatibility

        val file = try {
            // Option 1: Using app-specific external storage (works on most
            emulators)

            val appDir = getExternalFilesDir(null)
            val dir = File(appDir, "SDCardApp")

```

```

        if (!dir.exists()) {
            dir.mkdirs()
        }
        File(dir, "$registerNumber.txt")
    } catch (e: Exception) {
        // Option 2: Using legacy external storage as fallback
        val dir = File(Environment.getExternalStorageDirectory(),
"SDCardApp")

        if (!dir.exists()) {
            dir.mkdirs()
        }
        File(dir, "$registerNumber.txt")
    }

    // Create file
    val fileOutputStream = FileOutputStream(file)
    val data = "Register Number: $registerNumber\nName: $name\nCGPA:
$cgpa"

    fileOutputStream.write(data.toByteArray())
    fileOutputStream.close()

    Toast.makeText(this, "Data saved successfully to
${file.absolutePath}", Toast.LENGTH_LONG).show()

    // Clear fields after saving
    registerNumberEditText.text.clear()
    nameEditText.text.clear()
    cgpaEditText.text.clear()

} catch (e: IOException) {
    e.printStackTrace()
}

```



```

        Toast.makeText(this, "Error saving data: ${e.message}",
Toast.LENGTH_SHORT).show()

    }

}

private fun loadDataFromSD() {

    val registerNumber = registerNumberEditText.text.toString().trim()

    if (registerNumber.isEmpty()) {

        Toast.makeText(this, "Please enter a register number",
Toast.LENGTH_SHORT).show()

        return

    }

    try {

        // Try to find the file in multiple storage locations

        val file = findDataFile(registerNumber)

        if (file == null || !file.exists()) {

            Toast.makeText(this, "No data found for this register number",
Toast.LENGTH_SHORT).show()

            return

        }

        val fileContent = file.readText()

        val lines = fileContent.split("\n")

        // Parse the data

        for (line in lines) {

            when {

                line.startsWith("Name:") -> {

                    val name = line.substring(line.indexOf(":") + 1).trim()

                    nameEditText.setText(name)

```

```

        }

        line.startsWith("CGPA:") -> {

            val cgpa = line.substring(line.indexOf(":") + 1).trim()

            cgpaEditText.setText(cgpa)

        }

    }

    Toast.makeText(this, "Data loaded successfully from
    ${file.absolutePath}", Toast.LENGTH_SHORT).show()

    } catch (e: IOException) {

        e.printStackTrace()

        Toast.makeText(this, "Error loading data: ${e.message}",
        Toast.LENGTH_SHORT).show()

    }

}

private fun findDataFile(registerNumber: String): File? {

    // Try app-specific external storage first

    val appDir = getExternalFilesDir(null)

    val appSpecificDir = File(appDir, "SDCardApp")

    var file = File(appSpecificDir, "$registerNumber.txt")

    if (file.exists()) {

        return file

    }

    // Try legacy external storage

    try {

        val externalDir = File(Environment.getExternalStorageDirectory(),
        "SDCardApp")

        file = File(externalDir, "$registerNumber.txt")

    }

```

```

        if (file.exists()) {
            return file
        }
    } catch (e: Exception) {
        // Ignore and continue to other options
    }

    // Try cache directory as last resort
    val cacheDir = File(cacheDir, "SDCardApp")
    if (!cacheDir.exists()) {
        cacheDir.mkdirs()
    }
    file = File(cacheDir, "$registerNumber.txt")
    if (file.exists()) {
        return file
    }

    return null
}

// Checks if external storage is available for read and write
private fun isExternalStorageWritable(): Boolean {
    return Environment.getExternalStorageState() ==
Environment.MEDIA_MOUNTED
}

// Checks if external storage is available to at least read
private fun isExternalStorageReadable(): Boolean {
    return Environment.getExternalStorageState() in
        setOf(Environment.MEDIA_MOUNTED,
Environment.MEDIA_MOUNTED_READ_ONLY)
}

```

```

override fun onRequestPermissionsResult(
    requestCode: Int,
    permissions: Array<String>,
    grantResults: IntArray
) {
    super.onRequestPermissionsResult(requestCode, permissions,
grantResults)

    if (requestCode == STORAGE_PERMISSION_CODE) {
        if (grantResults.isNotEmpty() && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
            Toast.makeText(this, "Storage permission granted",
Toast.LENGTH_SHORT).show()
        } else {
            Toast.makeText(this, "Storage permission denied",
Toast.LENGTH_SHORT).show()
        }
    }
}
}

```

Output

The image displays two side-by-side screenshots of a mobile application interface titled "SD Card".

Left Screenshot: The interface has a purple header bar with the text "SD Card". Below the header, there are three white input fields. The first field contains the text "001". The second field contains the text "Aurthur Morgan". The third field contains the text "7.5". Below the input fields, there are two purple buttons: "Save" and "Load".

Right Screenshot: The interface has a purple header bar with the text "SD Card". Below the header, there are three white input fields. The first field contains the placeholder text "Enter the register number...". The second field contains the placeholder text "Enter the name...". The third field contains the placeholder text "Enter the CGPA...". Below the input fields, there are two purple buttons: "Save" and "Load". At the bottom of the screen, there is a light gray toast message with a green checkmark icon and the text "Data saved successfully to / storage/emulated/0/Android/data/co...".

Result:

The SD Card Access application successfully reads and writes data to the SD card or internal storage when tested on a mobile device with appropriate permissions.

